Activity 1.5 – 8-bit Ripple Carry Adder/Subtractor

1. Design Approach:

Designed an 8-bit ripple carry adder/subtractor using Verilog HDL. The circuit is constructed using full-adder modules connected through a generate loop. Subtraction is performed using 2's complement.

2. Verification Steps:

- Created a separate testbench.
- Tested addition with different input values.
- Tested subtraction with different input values.
- Checked zero and maximum input values.
- Verified carry and borrow conditions.
- Simulated using ModelSim.

3. Results:

The testbench compares the 8-bit result and carry output with the expected arithmetic result. PASS/FAIL messages are displayed.

4. Issues Faced:

Carry output must be interpreted correctly during subtraction. In 2's complement subtraction, carry-out indicates no borrow.

5. Conclusion:

Successfully designed and verified an 8-bit ripple carry adder/subtractor using full-adder modules.